DS_A1_Q1

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1 R code for creating column plot of mean fuel efficiency

To create a bar chart for average fuel efficiencies of city and highway grouped by car classes, we should group the data and calculate their each mean fuel efficiencies, as Table 1 shows.

```
# Library packages and load data
pacman::p_load(tidyverse, dplyr)
data(mpg, package ="ggplot2")

# Calculate the mean efficiency
mean_eff <- mpg %>%
  group_by(class) %>%
  summarise(cty = mean(cty),
            hwy = mean(hwy))

# Sort by city fuel efficiency and convert the tibble to long form
mean_eff <- mean_eff %>%
  arrange(cty) %>%
  mutate(class = factor(class, levels = class)) %>%
  gather(key = "measure", value = "mean", cty, hwy)

# Plot
ggplot(mean_eff, aes(x = class, y = mean, fill = measure)) +
  geom_col(position = "dodge", color = "black", linewidth = 0.5) +
  # Add title and axis title
  labs(
    title = "Column plot of mean fuel efficiency",
    x = "Vehicle class",
    y = "Mean fuel efficiency (miles per gallon)"
  ) +
  # Set filling colours and themes
  scale_fill_manual(values = c("cty" = "#0d0887", "hwy" = "#e16462")) +
  theme_bw() +
  theme(legend.position = "bottom")
```

An important aspect to highlight is that the bars in the plot should be ordered from lowest to highest based on the city's mean fuel efficiency. Therefore, we need to use R code to ensure the bars are plotted in the correct order.

Table 1: Mean Fuel Efficiencies grouped by Vehicle Classes.

class	measure	mean
pickup	cty	13.00000
suv	cty	13.50000
2seater	cty	15.40000
minivan	cty	15.81818
midsize	cty	18.75610
compact	cty	20.12766
subcompact	cty	20.37143
pickup	hwy	16.87879
suv	hwy	18.12903
2seater	hwy	24.80000
minivan	hwy	22.36364
midsize	hwy	27.29268
compact	hwy	28.29787
subcompact	hwy	28.14286

2 Generate the plot

- **The figure should have titles**
- **The colours filled in the bars**: #0d0887 for city and #e16462 for highway
- **The bars of highways** should be placed next to those of cities
- **The legend** should be placed at the bottom of the figure

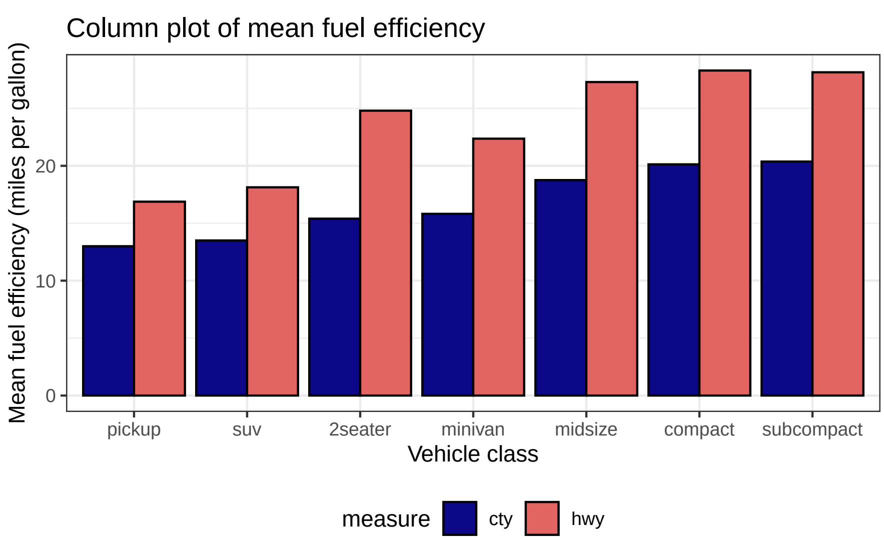

Figure 1: Column plot of mean fuel efficiency

So we would generate the Figure 1 as the question asked.